**Strain comparisons** % highly related strains (>=99% ANI) % less related strains (<99% ANI)

## Milk kefir

| | | |
|---|---|---|
| *Pseudomonas_E lundensis* | 100 | 0 |
| *Kluyvera cryocrescens* | 100 | 0 |
| *Pseudomonas_E sp002966775* | 95.5 | 4.5 |
| *Leuconostoc pseudomesenteroides_A* | 93.3 | 6.7 |
| *Leuconostoc pseudomesenteroides* | 93.3 | 6.7 |
| *Pseudomonas_E fragi* | 92.9 | 7.1 |
| *Bacillus_A mycoides* | 91.7 | 8.3 |
| *d__Bacteria;p__Firmicutes;c__Bacilli;o__Paenibacillales;f__Paenibacillaceae;g__Paenibacillus;s__* | 90.5 | 9.5 |
| *d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Enterobacterales;f__Enterobacteriaceae;g__Kluyvera;s__* | 87.5 | 12.5 |
| *Acinetobacter johnsonii* | 84.6 | 15.4 |
| *Lactococcus lactis_E* | 80.3 | 19.7 |
| *Escherichia flexneri* | 80 | 20 |
| *Raoultella terrigena* | 75 | 25 |
| *Pseudomonas_E proteolytica_A* | 75 | 25 |
| *Pseudomonas_E brenneri_B* | 75 | 25 |
| *Bacillus_A wiedmannii* | 75 | 25 |
| *Serratia liquefaciens* | 66.7 | 33.3 |
| *Paenibacillus sp001955855* | 66.7 | 33.3 |
| *Leclercia adecarboxylata* | 66.7 | 33.3 |
| *Lactococcus lactis* | 64.4 | 35.6 |
| *Pseudomonas_E putida* | 61.5 | 38.5 |
| *Lactococcus piscium_C* | 60.8 | 39.2 |
| *Pseudomonas_E helleri* | 54.5 | 45.5 |
| *Pantoea agglomerans* | 54.5 | 45.5 |
| *Citrobacter freundii* | 53.3 | 46.7 |
| *Acinetobacter albensis* | 53.3 | 46.7 |
| *Lelliottia amnigena* | 52.5 | 47.5 |
| *Thermus parvatiensis* | 48.4 | 51.6 |
| *Clostridium beijerinckii* | 48.3 | 51.7 |
| *Escherichia coli_D* | 40 | 60 |
| *Rahnella aquatilis_B* | 22.2 | 77.8 |
| *Pseudomonas_E fragi_B* | 21.4 | 78.6 |
| *Leuconostoc mesenteroides* | 18.1 | 81.9 |
| *Carnobacterium maltaromaticum* | 16.7 | 83.3 |
| *Acetobacter orientalis* | 14.3 | 85.7 |
| *Acetobacter fabarum* | 1.1 | 98.9 |
| *d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Pseudomonadales;f__Pseudomonadaceae;g__Pseudomonas_E;s__* | 0 | 100 |
| *d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Enterobacterales;f__Enterobacteriaceae;g__Superficieibacter;s__* | 0 | 100 |
| *d__Bacteria;p__Proteobacteria;c__Alphaproteobacteria;o__Acetobacterales;f__Acetobacteraceae;g__Acetobacter;s__* | 0 | 100 |
| *d__Bacteria;p__Firmicutes_A;c__Clostridia;o__Clostridiales;f__Clostridiaceae;g__Clostridium;s__* | 0 | 100 |
| *d__Bacteria;p__Firmicutes;c__Bacilli;o__Lactobacillales;f__Lactobacillaceae;g__Lentilactobacillus;s__* | 0 | 100 |
| *d__Bacteria;p__Firmicutes;c__Bacilli;o__Lactobacillales;f__Lactobacillaceae;g__Lactobacillus;s__* | 0 | 100 |
| *Streptococcus thermophilus* | 0 | 100 |
| *Streptococcus parauberis* | 0 | 100 |
| *Pseudomonas_E sp003852405* | 0 | 100 |
| *Pseudomonas_E sp001320045* | 0 | 100 |
| *Massilibacillus massiliensis* | 0 | 100 |
| *Macrococcus_B caseolyticus* | 0 | 100 |
| *Leuconostoc citreum* | 0 | 100 |
| *Lentilactobacillus parakefiri* | 0 | 100 |
| *Lentilactobacillus kefiri* | 0 | 100 |
| *Lactococcus raffinolactis* | 0 | 100 |
| *Lactobacillus kefiranofaciens* | 0 | 100 |
| *Lactobacillus helveticus* | 0 | 100 |
| *Klebsiella_A grimontii* | 0 | 100 |
| *Hafnia paralvei* | 0 | 100 |
| *Flavobacterium frigidarium* | 0 | 100 |
| *Enterococcus faecalis* | 0 | 100 |
| *Clostridium_B tyrobutyricum* | 0 | 100 |

0 25 50 75 100

## Water kefir

| | | |
|---|---|---|
| *Pseudomonas_E lactis* | 90.9 | 9.1 |
| *d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Enterobacterales;f__Enterobacteriaceae;g__Rahnella;s__* | 88.9 | 11.1 |
| *Gluconobacter oxydans_B* | 86 | 14 |
| *Agrobacterium pusense* | 80 | 20 |
| *Gluconobacter cerinus* | 76.2 | 23.8 |
| *Pseudomonas_E peli* | 75 | 25 |
| *d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Burkholderiales;f__Burkholderiaceae;g__Acidovorax_D;s__* | 69 | 31 |
| *d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Burkholderiales;f__Burkholderiaceae;g__Hydrogenophaga;s__* | 66.7 | 33.3 |
| *Aeromonas rivipollensis* | 66.7 | 33.3 |
| *Aeromonas encheleia* | 66.7 | 33.3 |
| *Acinetobacter guillouiae* | 66.7 | 33.3 |
| *Acidovorax_D sp005405905* | 66.7 | 33.3 |
| *Acidovorax_D sp001633105* | 66.7 | 33.3 |
| *Acetobacter persici* | 58.3 | 41.7 |
| *d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Burkholderiales;f__Burkholderiaceae;g__Aquabacterium;s__* | 58 | 42 |
| *Liquorilactobacillus ghanensis* | 55.6 | 44.4 |
| *Pantoea vagans* | 54.5 | 45.5 |
| *Gluconobacter albidus* | 48.9 | 51.1 |
| *Lacticaseibacillus paracasei* | 43.5 | 56.5 |
| *Pseudomonas_E veronii* | 42.9 | 57.1 |
| *Acetobacter indonesiensis* | 42.9 | 57.1 |
| *d__Bacteria;p__Actinobacteriota;c__Actinomycetia;o__Actinomycetales;f__Microbacteriaceae;g__Pseudoclavibacter_A;s__* | 42.1 | 57.9 |
| *d__Bacteria;p__Firmicutes;c__Bacilli;o__Lactobacillales;f__Lactobacillaceae;g__Lacticaseibacillus;s__* | 39.4 | 60.6 |
| *Acinetobacter ursingii* | 35.6 | 64.4 |
| *d__Bacteria;p__Proteobacteria;c__Alphaproteobacteria;o__Acetobacterales;f__Acetobacteraceae;g__Komagataeibacter;s__* | 30.3 | 69.7 |
| *d__Bacteria;p__Proteobacteria;c__Alphaproteobacteria;o__Acetobacterales;f__Acetobacteraceae;g__Gluconobacter;s__* | 30.1 | 69.9 |
| *d__Bacteria;p__Proteobacteria;c__Alphaproteobacteria;o__Rhizobiales;f__Rhizobiaceae;g__Allorhizobium;s__* | 25 | 75 |
| *Liquorilactobacillus mali_A* | 12.5 | 87.5 |
| *Acetobacter malorum* | 8.7 | 91.3 |
| *d__Bacteria;p__Firmicutes;c__Bacilli;o__Lactobacillales;f__Lactobacillaceae;g__Liquorilactobacillus;s__* | 5 | 95 |
| *d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Pseudomonadales;f__Moraxellaceae;g__Perlucidibaca;s__* | 0 | 100 |
| *d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Pseudomonadales;f__Moraxellaceae;g__Alkanindiges;s__* | 0 | 100 |
| *d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Enterobacterales;f__Aeromonadaceae;g__;s__* | 0 | 100 |
| *d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Burkholderiales;f__Burkholderiaceae;g__Pelomonas;s__* | 0 | 100 |
| *d__Bacteria;p__Proteobacteria;c__Alphaproteobacteria;o__Sphingomonadales;f__Sphingomonadaceae;g__Sphingomonas;s__* | 0 | 100 |
| *d__Bacteria;p__Proteobacteria;c__Alphaproteobacteria;o__Caulobacterales;f__Caulobacteraceae;g__Caulobacter;s__* | 0 | 100 |
| *d__Bacteria;p__Firmicutes;c__Bacilli;o__Bacillales;f__Sporolactobacillaceae;g__Sporolactobacillus;s__* | 0 | 100 |
| *d__Bacteria;p__Actinobacteriota;c__Actinomycetia;o__Actinomycetales;f__Bifidobacteriaceae;g__Bifidobacterium;s__* | 0 | 100 |
| *Zymomonas mobilis* | 0 | 100 |
| *Sphingomonas sp002292295* | 0 | 100 |
| *Sphingomonas aurantiaca* | 0 | 100 |
| *Sphingobium sp003609795* | 0 | 100 |
| *Schleiferilactobacillus perolens* | 0 | 100 |
| *Pseudomonas_E rhodesiae* | 0 | 100 |
| *Pseudescherichia vulneris* | 0 | 100 |
| *Oenococcus oeni* | 0 | 100 |
| *Microbacterium sp000411455* | 0 | 100 |
| *Microbacterium hominis* | 0 | 100 |
| *Methylobacterium aquaticum* | 0 | 100 |
| *Liquorilactobacillus satsumensis* | 0 | 100 |
| *Liquorilactobacillus nagelii* | 0 | 100 |
| *Leuconostoc gelidum* | 0 | 100 |
| *Lentilactobacillus hilgardii* | 0 | 100 |
| *Komagataeibacter hansenii* | 0 | 100 |
| *Comamonas testosteroni_B* | 0 | 100 |
| *Comamonas acidovorans* | 0 | 100 |
| *Chryseobacterium mucoviscidosis* | 0 | 100 |
| *Caulobacter sp002742645* | 0 | 100 |
| *Brochothrix thermosphacta* | 0 | 100 |
| *Brevundimonas mediterranea* | 0 | 100 |
| *Bifidobacterium psychraerophilum* | 0 | 100 |
| *Bifidobacterium aquikefiri* | 0 | 100 |
| *Agrobacterium sp900013535* | 0 | 100 |
| *Acetobacter syzygii* | 0 | 100 |
| *Acetobacter papayae* | 0 | 100 |
| *Pseudomonas_E lundensis* | 25 | 75 |
| *Pseudomonas_E sp002966775* | 63.6 | 36.4 |
| *Leuconostoc pseudomesenteroides_A* | 88 | 12 |
| *d__Bacteria;p__Firmicutes;c__Bacilli;o__Paenibacillales;f__Paenibacillaceae;g__Paenibacillus;s__* | 85.7 | 14.3 |
| *d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Enterobacterales;f__Enterobacteriaceae;g__Kluyvera;s__* | 87.5 | 12.5 |
| *Acinetobacter johnsonii* | 90.8 | 9.2 |
| *Lactococcus lactis_E* | 73.6 | 26.4 |
| *Raoultella terrigena* | 37.5 | 62.5 |
| *Leclercia adecarboxylata* | 66.7 | 33.3 |
| *Lactococcus lactis* | 87.1 | 12.9 |
| *Pseudomonas_E putida* | 61.5 | 38.5 |
| *Pseudomonas_E helleri* | 57.6 | 42.4 |
| *Lelliottia amnigena* | 60 | 40 |
| *Clostridium beijerinckii* | 91.7 | 8.3 |
| *Rahnella aquatilis_B* | 22.2 | 77.8 |
| *Pseudomonas_E fragi_B* | 21.4 | 78.6 |
| *Carnobacterium maltaromaticum* | 3.3 | 96.7 |
| *Acetobacter orientalis* | 89.3 | 10.7 |
| *Acetobacter fabarum* | 1.1 | 98.9 |
| *d__Bacteria;p__Proteobacteria;c__Gammaproteobacteria;o__Pseudomonadales;f__Pseudomonadaceae;g__Pseudomonas_E;s__* | 0 | 100 |
| *d__Bacteria;p__Proteobacteria;c__Alphaproteobacteria;o__Acetobacterales;f__Acetobacteraceae;g__Acetobacter;s__* | 18.8 | 81.2 |
| *Leuconostoc citreum* | 0 | 100 |
| *Lactobacillus kefiranofaciens* | 0 | 100 |
| *Klebsiella_A grimontii* | 0 | 100 |

0 25 50 75 100

Pairwise comparisons (%)